Three-dimensional atom localisation with calibrated uncertainty from multislice electron ptychography: mapping polar order in a $PbTiO_3/SrTiO_3$ superlattice

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Polar vortices in ferroelectric superlattices are three-dimensional: the polarisation rotates along the electron beam as well as across it, so a projection image integrates away the structure it is used to measure. Multislice electron ptychography recovers depth from a single orientation. Converting its phase volume into atomic coordinates is obstructed, however, by a highly anisotropic response of 0.1 Å in-plane against 1–2 Å along the beam, which buries the oxygen sublattice beneath its titanium neighbours. 4D-STEM of a relaxed $PbTiO_3/SrTiO_3$ supercell was simulated, inverted, and every atom located by fitting the volume as a superposition of a measured single-atom response. The method recovers 95.5% of bulk oxygen with 1.1% species confusion, whereas three-dimensional deconvolution followed by peak detection reaches at most 68% over the same band. That advantage is concentrated entirely in the axially overlapped oxygen, for which recall is 82% against 9–26%. Per-atom uncertainties combine a joint Cramér–Rao bound with Mondrian conformal calibration, the former correcting the twofold variance inflation a per-peak bound misses at the 1.95 Å Ti–O spacing. The resulting intervals hold their nominal coverage to within 1% and transfer without retuning to an independent reconstruction. Propagated through the Ti–O_6 off-centring, they yield the local polarisation with error bars. The in-plane component is recovered to a median 0.007 Å and 0.9°, while the along-beam component is not determined, and that conclusion is reached without reference to ground truth. Core-hole DFT indicates the missing component remains spectroscopically accessible, the apical-oxygen O K edge being 78% dichroic to the polar-axis orientation with a monotonic 19% arising from the displacement itself, although weighted over both oxygen sites the full edge retains 37% of that dichroism. Recovering it would nonetheless require a collection aperture $\lesssim 2$ mrad rather than the 100 mrad ptychography optics.

I. INTRODUCTION

Ferroelectric oxide superlattices host polar textures such as vortices, skyrmions and labyrinths that have no counterpart in bulk ferroelectrics [1–4]. In $PbTiO_3/SrTiO_3$ (PTO/STO), competition between elastic, electrostatic and gradient energies drives the local dipoles into rotational arrangements whose topology, rather than magnitude alone, is the property of interest. Establishing that topology requires the measurement of a vector field at atomic resolution. The standard tool is atomic-resolution STEM, in which the polarisation is inferred from the sub-ångström offset of the B-site cation from the centre of its oxygen cage, mapped column by column.

Such a measurement is a projection, and returns only the component of the off-centring lying in the image plane. For a uniformly polarised domain this is minor and can be lifted by imaging a second zone axis, but for a vortex it cannot, because the polarisation rotates with depth. Different depths within one column then carry different, sometimes opposing, dipoles, of which the image reports the average. The structure defining the texture is consequently the structure the measurement removes.

Multislice electron ptychography (MEP) provides an alternative, reconstructing the specimen as a stack of slices and thereby achieving optical sectioning from a single orientation [5–7]. Its lateral resolution is set by the illumination angle rather than by the lens [8–10], and it has been applied to dopant depths in bulk crystals [6], oxygen vacancy sites in nickelates [11] and, with a few coupled small-angle projections, to sub-nanometre depth resolution [12]. Atomic electron tomography offers quantified three-dimensional coordinates [13–15] but requires a large tilt series, which is prohibitive for a beam-sensitive oxide.

The difficulty lies in what follows the reconstruction. A single-orientation measurement is aperture-limited in depth, the lateral and axial resolutions scaling as $\lambda/2\alpha$ and λ/α^2, which for the 300 keV, 100 mrad probe used here are ≈ 0.10 Å and ≈ 2.0 Å. Since the axial extent exceeds the 1.95 Å Ti–O spacing, an oxygen atom lies within the axial flank of its titanium neighbour and local peak detection cannot separate the two. The established remedy of three-dimensional Richardson–Lucy or maximum-entropy deconvolution followed by peak analysis [16] was developed for sparse features rather than for a dense sublattice. The accepted two-dimensional approach, model-based fitting of a peak superposition [17], does handle overlap explicitly, but it has not been extended to three dimensions for every atom of a dense lattice simultaneously.

A second difficulty is more consequential for any physical conclusion. A polarisation map is a difference of coordinates, each carrying an error, so whether a measured off-centring is real cannot be judged without a per-atom uncertainty that is calibrated rather than merely reported. Such uncertainties are rarely provided: a leastsquares covariance is optimistic on near-noiseless data, and the per-peak Cramér–Rao bound assumes all neighbours known exactly, which is the assumption that fails on an overlapped lattice.

This work addresses both difficulties. 4D-STEM of a relaxed PTO/STO supercell was simulated and reconstructed by MEP, and every atom was then located by fitting the phase volume as a superposition of a measured single-atom response, which extends model-based quantification [17] to three dimensions and to all species simultaneously. Per-atom uncertainties combining a joint Cramér–Rao bound with conformal calibration were attached to those coordinates, shown to transfer to an independent reconstruction, and propagated into a Ti–O_6 polarisation map. Core-hole DFT was then used to assess whether the component inaccessible to ptychography is recoverable spectroscopically. Throughout, the localisation has no access to the ground truth, which is used only to score the result.

II. METHODS

A. Model system

The test object was a relaxed atomistic supercell of a PTO/STO superlattice provided by collaborators, of dimensions $70.0 \times 70.0 \times 47.9$ Å and containing 19 440 atoms, comprising 1944 formula units of each material. Both are perovskites (ABO_3), with heavy Pb or Sr on the A site, Ti on the B site and oxygen on an octahedral cage; in the PTO layers the B-site titanium and A-site lead are displaced from their centrosymmetric positions and the resulting dipoles order into a labyrinth pattern. The local polarisation was quantified throughout by the B-site off-centring

$$\boldsymbol{\delta}_i \;=\; \mathbf{r}_{\mathrm{Ti},i} \;-\; \tfrac{1}{6} \sum_{j\in \mathrm{O}_6(i)} \mathbf{r}_{\mathrm{O},j}, \tag{1}$$

the displacement of each titanium from the charge centre of its six nearest oxygens, mapped in Fig. 1(a).

The system is an exacting test of localisation for three reasons. The scattering contrast spans two orders of magnitude, from lead ($Z = 82$) to oxygen ($Z = 8$), placing the light sublattice near the noise floor, and the signal of interest is a sub-ångström displacement of atoms 1.9–3.9 Å apart. The polarisation, moreover, twists with depth into a vortex tube within the scanned volume (Fig. 1c). The beam was set along a pseudocubic $\langle 100\rangle$ direction, giving a ≈ 70 Å path.

B. 4D-STEM simulation

The simulation implements the forward operator $\mathcal{S}$ mapping the structure to the data a microscope would record. A convergent probe of semi-angle $\alpha = 100$ mrad at 300 keV ($\lambda = 1.969$ pm) was focused 20 Å above the entrance surface. It was then propagated through the specimen by the multislice recursion [18, 19]

$$\psi_{n+1} = \mathcal{P}_{\Delta z} \otimes \big[\, t_n(\mathbf{r})\, \psi_n(\mathbf{r}) \,\big], \qquad t_n = \exp\big[\, i\, \sigma\, v_n(\mathbf{r}) \,\big], \tag{2}$$

in which v_n is the projected potential of slice n, σ the interaction constant and $\mathcal{P}_{\Delta z}$ the Fresnel propagator; the recorded quantity is $I(\mathbf{r}_p, \mathbf{k}) = |\mathcal{F}\psi_N|^2$. Calculations used abTEM [20] on GPU with Lobato potentials [21]. Thermal motion was included in the frozen-phonon approximation [22] as an incoherent average over 16 configurations, with amplitudes set per species from tabulated Debye–Waller factors [23]. Lead and oxygen vibrate roughly twice as far as titanium ($B = 0.90$ and 0.80 against 0.45 Å^2), so a single averaged amplitude would misweight the diffuse background of the species hardest to detect. A finite electron budget was imposed as shot noise, $n \sim \mathrm{Poisson}(N_e \tilde{I})$, over $D = 10^4$–10^{10} e Å^{-2}.

C. Multislice ptychographic reconstruction

Reconstruction inverts $\mathcal{S}$ by iterative ptychography [24, 25], the single multiplicative projection being replaced by a stack of N_L transmission slices [7]. Slices and probe were recovered jointly by minimising the negative log-likelihood of the shot-noise model, implemented as the variance-stabilised amplitude metric [26] in the LSQ-ML engine of Odstrčil *et al.* [27] within PtychoShelves [25]. Reconstructions used $N_L = 70$ slices (coherent, $\Delta z_{\mathrm{rec}} \approx 1.0$ Å) or 105 (thermally averaged, 0.67 Å). No regularisation was applied to the inter-layer coupling, so that the axial structure is set by the data rather than by a smoothness prior, which would bias the coordinate subsequently quantified. For the dosed data the illumination was treated as a mixed state [28] with orthogonal probe relaxation [29], allowing the thermal diffuse background to be absorbed by incoherent degrees of freedom rather than forced into the coherent object. Parameters are collected in Table A1 of Appendix A.

D. Atom localisation

The reconstructed object was treated as a sparse superposition of identical single-atom responses,

$$\phi(\mathbf{r}) = \sum_{i=1}^{N} \beta_i\, K(\mathbf{r} - \mathbf{r}_i) + b(\mathbf{r}) + \varepsilon(\mathbf{r}), \qquad \beta_i \geq 0, \tag{3}$$

in which ϕ is the per-layer median-subtracted phase, $(\beta_i, \mathbf{r}_i)$ the amplitude and three-dimensional position of atom i, and b a slowly varying background removed by a high-pass filter. Since the reconstruction is a nonlinear inverse problem, Eq. (3) is an empirical linearisation rather than a derived identity, and is tested directly in Sec. III A.

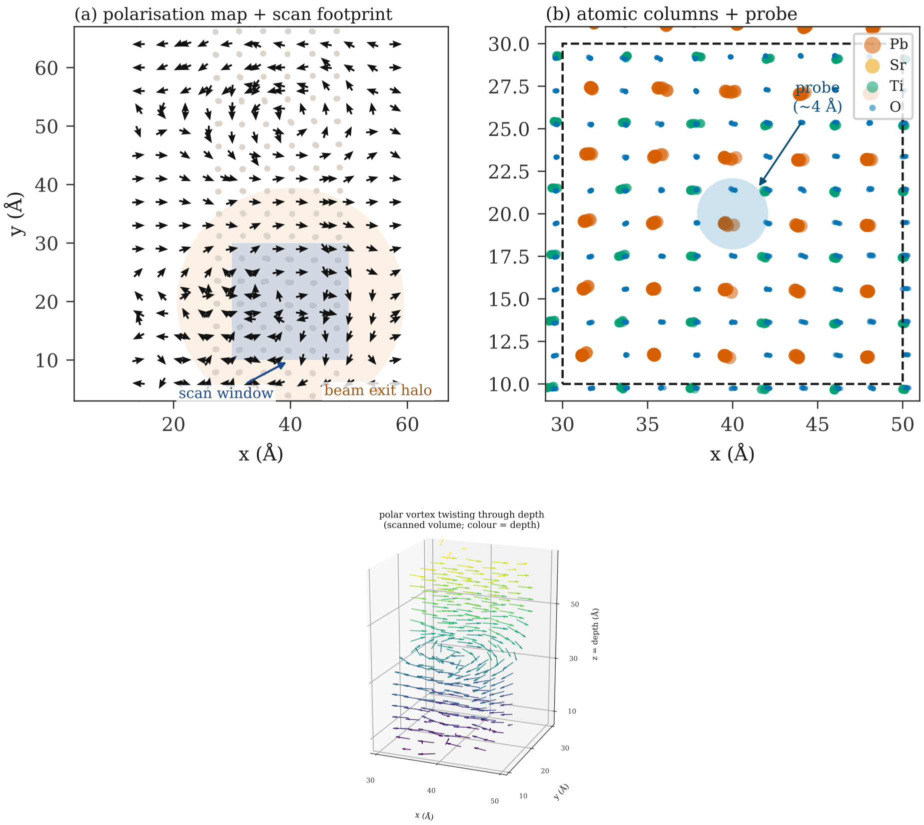

FIG. 1. The model system. (a) In-plane polarisation over the whole cell viewed down the beam, with one mean-direction arrow per atomic column, resolving the polar domains and vortices. The shaded square indicates the 20 Å scan window and the shaded disc the exit-wave footprint spanned by the recorded diffraction, which is why the full box was simulated. (b) Atomic columns in beam projection within the scan region, with the overfocused probe (≈ 4 Å) overlaid; the lead columns split into polar dumbbells. (c) Polarisation of the scanned volume in three dimensions, with the local dipoles (arrows, coloured by depth) twisting into a vortex tube along the beam. This depth variation is integrated away by a projection.

The kernel K was measured rather than parameterised, a single isolated atom being propagated through the byte-identical simulation and reconstruction pipeline and its reconstructed response taken as K. This is necessary because K is strongly anisotropic with an asymmetric axial profile, so no simple parametric form is faithful. The lead kernel was used for all species, since an isolated oxygen or titanium does not reconstruct above the noise floor; the response shape belongs to the imaging system whereas the amplitude β carries Z.

Estimation proceeds in two stages. The support is selected by matching pursuit on the dictionary $\{K(\cdot - \mathbf{r})\}$ [30, 31], iteration being stopped at a detection floor set relative to the local matched-filter noise rather than to an absolute phase value, so that the threshold transfers between reconstructions of differing amplitude. The support is then refined by block-coordinate descent, each atom being updated by Gauss–Newton steps on the local residual with the current models of all other atoms subtracted, so that overlap is accounted for to first order at every update. Candidates are retained on a shape-based goodness-of-fit criterion rather than on amplitude, which preserves weak but well-formed responses. The procedure extends model-based fitting of a peak superposition [17] to three dimensions with a measured kernel, applying to every atom of a dense lattice a depth-from-axial-response estimate established for isolated dopants [6, 32]. Columns are then typed from their own fitted-amplitude statistics, and sites predicted by a column's own periodicity but left empty by selection are re-examined at a reduced threshold, such lattice-constrained detections being flagged so that the constrained and unconstrained populations are reported separately. Thresholds, gates and the full stage-by-stage specification are given in Appendix B.

E. Uncertainty quantification

A model standard error, computed without ground truth, was separated from a calibration step converting it into intervals with a stated coverage.

The model error has two terms, $\sigma^2 = \sigma^2_{\mathrm{stat}} + \sigma^2_{\mathrm{sys}}$. The statistical term is the Cramér–Rao bound of the *joint* estimation problem, obtained by assembling the Jacobian over all atoms in a subvolume and inverting the Fisher matrix once, rather than the per-peak bound used to predict single-dopant precision [6, 17]. On a dense sublattice the per-peak form is the bound obtained with all neighbours known exactly, and is therefore optimistic by the variance inflation factor of the overlapping templates. The systematic term $\sigma_{\rm sys}$ is the position spread induced by kernel mismatch, which requires no ground truth and so transfers to experimental data. Intervals were then obtained by split-conformal calibration [33, 34] in the Mondrian (label-conditional) variant [35], stratified by species, detection mode and depth band, so that coverage holds per stratum rather than only on average. The 95% interval is the default in released coordinates, and the calibration table and per-stratum coverage are exported so that a dataset on which the calibration fails to transfer is identified rather than trusted silently. The derivation, the measured inflation factors and the coverage tables are given in Appendix C.

F. Polarisation from located atoms

Equation (1) was evaluated on the located atoms exactly as on the model, the six nearest located oxygens within 2.8 Å forming the cage. Titanium whose cage is incomplete in the reconstruction was excluded and reported separately rather than completed from the lattice. Uncertainty on $\boldsymbol{\delta}$ was propagated by Monte Carlo, using 400 draws, from the per-atom conformal half-widths of Sec. II E, that is from quantities available without ground truth.

G. First-principles EELS

The energy-loss near-edge structure (ELNES) is governed by the dynamic form factor entering the double-differential cross-section [36]. For the small momentum transfer $\mathbf{q}$ of a near-axis collection, the dipole expansion gives

$$S(\mathbf{q},E) \propto q^2 \sum_f \left|\langle f|\hat{\mathbf{q}}\cdot\mathbf{r}|i\rangle\right|^2 \delta(E-E_f+E_i), \qquad (4)$$

so that the ELNES measures the symmetry-projected unoccupied local density of states along $\hat{\mathbf{q}}$, which for a small on-axis aperture is essentially the beam direction, and in an anisotropic crystal depends on its orientation relative to the polar axis [37]. The form of this expression frames the whole study. Since $|\langle f|\hat{\mathbf{q}}\cdot\mathbf{r}|i\rangle|^2$ is even in $\mathbf{q}$, the dipole ELNES is sensitive to the magnitude of the along-beam polarisation but not to its sign. A real measurement, moreover, integrates over the convergence and collection apertures, and the anisotropic term is weighted by a geometric factor vanishing at the magic angle $\beta^\star \approx 4\theta_E$, with $\theta_E = E/(\gamma m_0 v^2)$ [38].

Equation (4) was evaluated with CASTEP [39] using the core-hole method [40], in which the ionised atom carries an on-the-fly-generated pseudopotential with one fewer core electron and a neutralising background, within a $2\times2\times2$ supercell that dilutes the hole from its periodic images. The core-to-conduction matrix elements were contracted by OptaDOS [41, 42] into the core-loss spectrum with $\hat{\mathbf{q}}$ set explicitly. The central quantity is the ELNES dichroism

$$\Delta(E) = S(\mathbf{q}\parallel c,E) - S(\mathbf{q}\perp c,E), \qquad (5)$$

which by reciprocity is the change in the measured edge when the polarisation rotates from along the beam to in-plane. The O K edge was taken as the primary observable, being an established probe of ferroelectric off-centring [43, 44], while Ti $L_{2,3}$ was treated for anisotropy trends only, its $2p$ multiplet structure not being captured by single-particle DFT [45]. Calculations ran on the Warwick Blythe cluster, each core-hole SCF requiring 40–45 min on 32 cores and each OptaDOS pass 75–90 min.

III. RESULTS AND DISCUSSION

A. The measured response and the forward model

The reconstructed single-atom response (Fig. 2a) has a full width at half maximum of 0.10 Å in-plane against 1.0 Å along the beam for the coherent reconstruction and 2.0 Å for the thermally averaged one. Because that axial width is comparable to or larger than the 1.95 Å Ti–O spacing, neighbouring responses overlap and the design is not separable. It is therefore necessary to establish whether the apparent blur represents overlap of a fixed response or a genuine loss of information. Equation (3) was tested by fixing $\{\mathbf{r}_i\}$ at the known positions and solving for $\{\beta_i\}$ alone. A comb of measured kernels at the true depths reproduces the observed depth profile of a mixed B–O column at a median $r = 0.94$ over 20 columns (Fig. 2b), and $r = 0.92$ in the full depth–x cross-section. The blur is consequently kernel overlap rather than lost information, which is why a method that models the overlap can outperform one that does not.

B. Localisation performance

Table I scores the method against the known structure, alongside the established alternatives run on the same volume with the same measured kernel, each at its own best-effort threshold. Over the bulk (z = 10–56 Å, excluding the entrance and exit artefact bands) the model fit recovers 95.5% of oxygen, 96.4% of titanium and 95.6% of lead, with 1.1% of species labels incorrect, an in-plane RMS accuracy of 0.032 Å and a depth RMS of 0.37 Å.

The aggregate oxygen figures understate what is being tested, so the sublattice was split by geometry. Oxygen

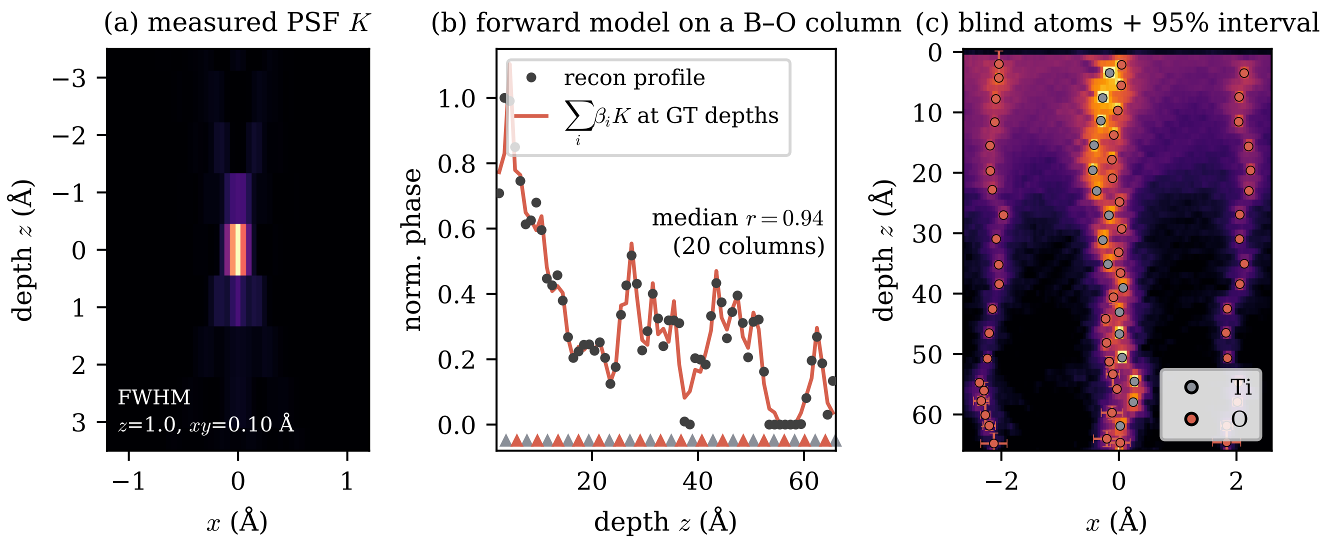

FIG. 2. Model-based localisation. (a) The measured point-spread function K in a depth–x slice, tight in-plane and elongated along the beam by the aperture-limited axial resolution of a single-orientation measurement. (b) Direct test of the forward model, Eq. (3): the measured depth profile of a mixed Ti–O column against the superposition $\sum_i \beta_i K$ of the measured response at the known atomic depths, matching at a median $r = 0.94$ over 20 columns. Triangles mark the true Ti (grey) and O (red) depths. The apparent 2–4 Å axial blur is overlap of a fixed response rather than a loss of information. (c) Blind atoms located on a raw cross-section, coloured by assigned species, with calibrated 95% intervals.

	recall (%)			oxygen split (%)	
method	Pb	Ti	O	isolated	overlapped
3-D peak-pick, raw	93	88	72	91	26
RL deconv. + peak-pick	93	87	58	79	9
MEM deconv. + peak-pick	93	89	59	79	13
1-D per-column deconv.	91	65	49	62	20
3-D model fit	88	89	89	92	82
unconstrained only	88	89	86	92	71
bulk, z = 10–56 Å	95.6	96.4	95.5	—	—

TABLE I. Recovery of each sublattice, whole field unless stated, on the same reconstruction with the same measured kernel. Isolated oxygen occupies its own column; overlapped oxygen shares a column with titanium 1.95 Å away. Precision is 0.97 for the model fit (0.98 unconstrained) against 0.84–0.99 for the baselines. Only the model fit returns species labels and per-atom uncertainties.

that is in-plane isolated, occupying its own column, is found by every method at 79–92%. Oxygen that is axially overlapped, sharing a column with titanium 1.95 Å away, is the difficult case, and it is here that the methods separate: 82% for the model fit against 26% for raw peak-picking, 13% for maximum-entropy deconvolution and 9% for Richardson–Lucy. This population accounts for the whole of the advantage.

Neither of the remaining comparisons behaved as anticipated. Deconvolution before detection does not assist, and Richardson–Lucy is actively detrimental to the oxygen, both routines losing oxygen relative to peak-picking the raw volume at 58–59% against 72%. They redistribute dynamic range towards the strongly scattering species, so weak oxygen falls below any relative floor, and they cannot restore the axial frequencies a single-orientation measurement never records, which are precisely those separating an oxygen atom from the flank of its titanium neighbour. Ishizuka *et al.* report maximum entropy as the stronger routine [16]; no advantage was reproduced on a dense lattice, so the limitation appears structural rather than a tuning artefact. The second surprise is that on noiseless data a good peak-picker is competitive on the heavy atoms, a raw local maximum being already a clean estimator of an isolated bright column. The margin quoted here is consequently a lower bound. It is tempting to expect the margin to widen at finite dose; Sec. III D shows that this cannot be tested at a single operating point, because the baseline saturates its detection limit there. A peak-picker cannot, at any dose, provide a species label or a calibrated error bar.

C. Calibrated uncertainty

Figure 3(a) quantifies why a per-peak bound is inadequate here. The variance inflation factor of the joint fit is 1.04 at the 3.9 Å heavy-atom spacing, where atoms are effectively independent, but 2.02 in depth and 2.28 in amplitude at the 1.95 Å Ti–O spacing, so that a per-peak bound understates σ_z by $\approx 1.4\times$ for every apical oxygen. Since 85% of atoms have a neighbour within 2.5 Å, the per-peak form is optimistic almost everywhere, and in-

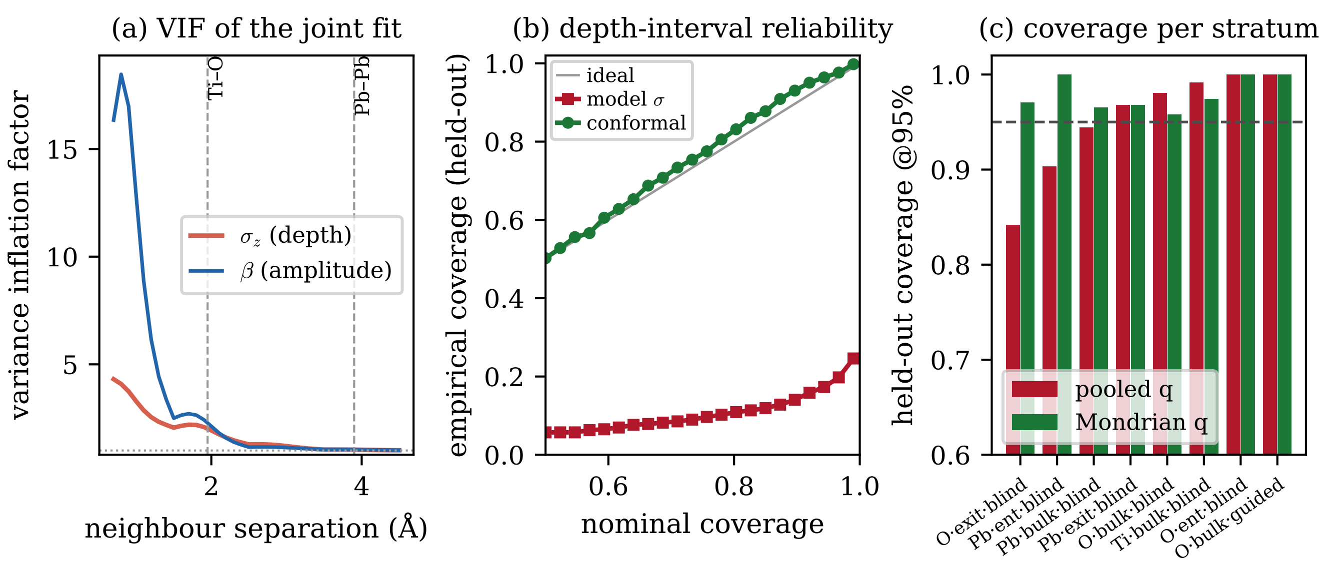

FIG. 3. Calibrated uncertainty. (a) Variance inflation factor of the joint fit against neighbour separation: near unity at the 3.9 Å heavy-atom spacing but ≈2 in depth, and larger in amplitude, at the 1.95 Å Ti–O spacing, so that a per-peak Cramér–Rao bound is optimistic wherever responses overlap. (b) Held-out reliability of the depth interval: the raw model standard error treated as Gaussian (red) severely under-covers on near-noiseless data, whereas the conformal interval (green) tracks the ideal diagonal. (c) Held-out coverage at the 95% level for the strata in which a single pooled quantile (red) departs most from target; the label-conditional quantile (green) restores coverage in each. Split-conformal, 50/50 calibration/test split.

verting the full Fisher matrix restores the inflation per atom without a tuned constant.

The raw model σ treated as Gaussian is nonetheless badly miscalibrated on near-noiseless simulation, covering only ≈ 8% of atoms at a nominal 68% (Fig. 3b), since the residual-based variance is near zero while the true error is dominated by systematics the residual cannot detect. Conformal calibration supplies the missing scale empirically: held out on a 50/50 split, the interval tracks the ideal diagonal across the whole range of nominal coverage, reaching 69/69/69% in (x, y, z) at a 68% target and 96/96/96% at 95%. The Mondrian stratification is not cosmetic, the per-stratum quantiles $q_{68}(z)$ ranging from 6.7 to 14.4, a factor of 2.2, so that a single pooled constant leaves some strata conservative and others optimistic. In Fig. 3(c) pooling drops the worst stratum to 84% against a 95% target, whereas the label-conditional quantile restores every stratum to 95–100%.

The decisive test is whether an uncertainty estimate survives a change of dataset. The identical code, with no tuned constants, was applied to an independent reconstruction of the same material at a different in-plane sampling (0.10 against 0.15 Å), a different slice discretisation (105 against 70 layers), roughly half the phase amplitude and real reconstruction noise. Re-running the conformal step on that volume's own atoms gives 69/69/69% at the 68% target and 96/96/96% at 95%. For comparison, the hand-tuned per-species error floors used in an earlier version of this work, calibrated to 68% coverage on the first volume, transferred to 38/47/56% on the second. The per-stratum quantiles on the harder volume span 10.9 to 23.3, which is the regime a single floor cannot serve.

The scope of these intervals needs stating carefully. Each is conditional on the atom having been correctly detected and typed, so that a spurious detection or species swap is a distinct failure mode, reported separately as precision (0.97) and confusion (1.1%). The exported species posterior is a within-column amplitude confidence and is not itself calibrated, although labels below 0.9 are enriched for mislabelling by more than an order of magnitude over the base rate. Finally, conformal calibration assumes exchangeability, which is why the workflow recalibrates per volume and exports the per-stratum coverage.

D. Behaviour at finite dose

Table II applies the same configuration over a dose series on the independent reconstruction. Performance is maintained to $10^8\,\mathrm{e\,Å^{-2}}$ and collapses below 10^6. The residual gap to the coherent reference (Ti 74% against 96% on the same bulk band) is attributable not to the localisation constants but to a harder reconstruction, having a 2.0 Å axial kernel against 1.0 Å and a depth-registration correlation of 0.44 against 0.63.

The baselines of Sec. III B were run over the same series, expecting the oxygen margin to widen once noise was present. It does not, for an instructive reason: peak detection is subject to a cap on the number of maxima

dose ($e\,Å^{-2}$)	found	precision	Pb	Ti	O	confusion
10^{10}	1575	0.85	89 %	74 %	60 %	12.4 %
10^{8}	1593	0.88	89 %	77 %	66 %	11.3 %
10^{6}	521	0.96	76 %	26 %	5 %	37.8 %
10^{4}	0	—	—	—	—	—

TABLE II. Dose series on an independent reconstruction (105 layers, 0.10 Å sampling), run with the same configuration as the coherent reference. Recalls are over the bulk band. The 10^6 row illustrates the uninformativeness of precision, which reads 0.96 while oxygen recall is 5%. At 10^4 the method returns no atoms at all rather than reporting noise.

returned, and it reaches that cap at every dose, returning the same 2500 detections at 10^{10} as at 10^4. Recall is then set by the cap rather than by the data, and the ordering becomes impossible, axially overlapped oxygen scoring 71–77% against 45–49% for the easier isolated oxygen. Counting true detections shows why: at 10^8 the model fit makes 1404 from 1593 returned and peak-picking 1407 from 2500, buying the same true detections with 1093 false ones. Discriminating at finite dose would need the baselines re-thresholded to a matched count, or a precision-recall curve rather than one operating point, so no claim is made from it here. What the series does establish is the value of abstention: at 10^4 the model fit returns nothing, whereas maximum-entropy deconvolution returns 3803 detections at a precision of 0.34.

The series carries a further lesson that outlives this dataset. Precision is not a safety indicator: at $10^6\,e\,Å^{-2}$ it reads 0.96 while titanium recall is 26%, oxygen 5% and 38% of labels are wrong, because it measures only the surviving bright lead atoms, which are correctly placed. Species confusion and σ-coverage are the diagnostics that respond, and both are printed as health warnings. Species assignment is coupled to depth registration as well, since a 1 Å depth error on a lattice whose Ti and O alternate every 1.95 Å swaps labels wholesale: correcting the offset from 1.26 to 0.29 Å raised titanium recall from 53% to 74% and reduced confusion from 19.0% to 12.4% at $10^{10}\,e\,Å^{-2}$. Sub-ångström depth accuracy is therefore a prerequisite for correct chemistry.

E. Polarisation with error bars

Evaluating Eq. (1) on the located atoms, 72% of bulk titanium acquire a complete oxygen cage from blindly located oxygens alone, of which 171 match a ground-truth titanium and are scored in Fig. 4.

The in-plane off-centring is recovered essentially exactly. The median error in δ_x is 0.005 Å and in δ_y 0.004 Å, against a true in-plane off-centring of mean magnitude 0.274 Å, for which the recovered mean magnitude is 0.273 Å. As a vector it has a median error of 0.007 Å and a 90th percentile of 0.017 Å, while its direction, the quantity from which a vortex map is constructed, has a median error of 0.9°, with 95% of titanium within 30°. The correlation between recovered and true δ_x is 0.96 over all matched atoms and 1.00 once a small tail is excluded.

That tail requires comment. Five per cent of titanium have a cage of the wrong composition, since a missing or misassigned oxygen displaces the centroid, and for these the in-plane error rises to a median of 0.45 Å, comparable to the signal itself. The propagated interval covers 99% of the well-caged titanium but only 12% of the tail. This is not a failure of the calibration but a statement of its scope: the interval is conditional on correct detection, whereas a gross error in cage composition is a detection failure and must be reported through precision, confusion and cage completeness rather than absorbed into a positional error bar. A practical map should carry both, and the 28% of titanium with incomplete cages should be shown as absent rather than interpolated.

The along-beam component behaves in the opposite manner, and this is the more consequential result. Its true spread across the scanned volume is small, $\sigma = 0.074$ Å, since in this zone axis the vortex polarisation is predominantly in-plane, whereas the propagated uncertainty on δ_z is 0.237 Å, more than three times the entire signal, and the correlation with truth is correspondingly poor at $r = 0.42$ with a median error of 0.122 Å. The along-beam component is therefore not measured by this experiment. The calibrated uncertainty establishes this without ground truth, since comparing the propagated σ_z with the spread of the recovered δ_z requires no reference structure; it declares the along-beam map uninformative while certifying the in-plane map to 0.01 Å. Propagated intervals cover 95/93/99% of true errors in (x, y, z) at a 95% target, so neither the confidence in-plane nor the pessimism along the beam is overstated.

Depth localisation of individual atoms is itself good, at 0.37 Å RMS, comfortably better than the ≈ 2 Å depth resolution, in the manner that single-molecule localisation improves upon the diffraction limit [46]. A polarisation is, however, a difference of two such coordinates, and differencing two 0.3 Å depth errors cannot resolve a 0.07 Å displacement. In-plane the error is 0.03 Å against a 0.27 Å signal, so the map is quantitative.

F. Can EELS see the missing component?

The second phase examines whether the along-beam polarisation leaves a spectroscopic signature, and was constructed as a validated staircase. CASTEP/PBE reproduces the ferroelectric instability: single-point energies across a displacement series $\mathbf{u} = s\,\mathbf{u}_0$ at fixed tetragonal strain place the fully polar structure 203 meV per formula unit below the centrosymmetric reference, monotonically, with the descent flattening towards $s = 1$ so that the experimental displacement lies at the PBE minimum. The core-hole recipe was fixed by reproducing the rutile TiO_2 O K ELNES, obtained with its textbook

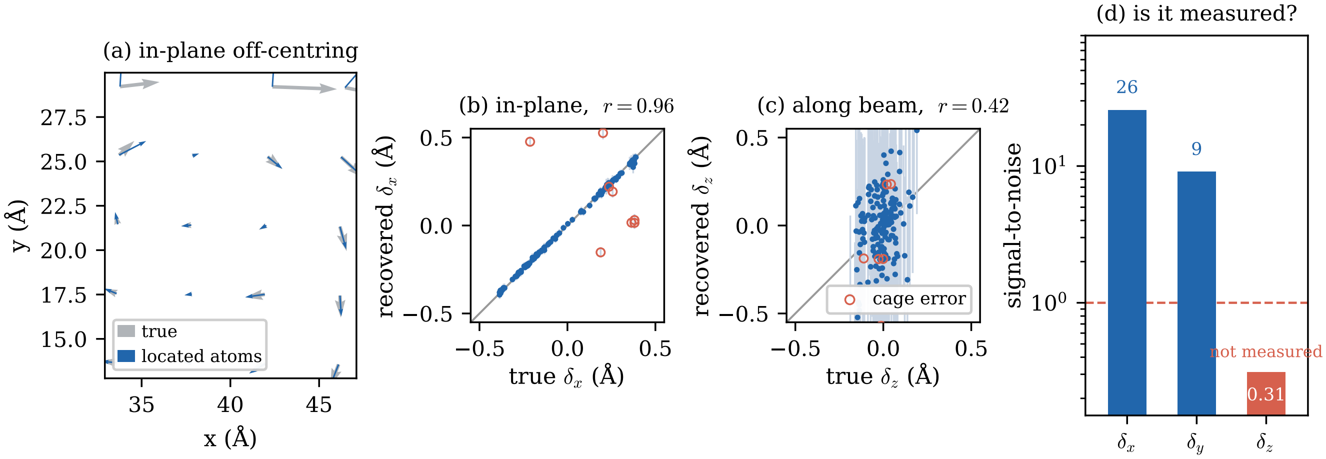

FIG. 4. Local polarisation from the located atoms. (a) In-plane $Ti–O_6$ off-centring averaged per atomic column: recovered (blue) over ground truth (grey). (b) Recovered against true in-plane off-centring, with propagated 95% intervals; open red circles mark the 5% of titanium whose oxygen cage has the wrong composition. (c) The same for the along-beam component. (d) Signal-to-noise per component, defined as the true spread of the component divided by the propagated σ on it, on a logarithmic axis with the decision line at unity. In-plane the uncertainty is a small fraction of the signal, whereas along the beam it exceeds it, which is a ground-truth-free statement that the along-beam polarisation is not determined by this measurement.

shape, the t_{2g}/e_g near-edge peaks split by 2.7 eV (experiment 2.5–3 eV) with $e_g > t_{2g}$. As an artefact check the dichroism of the cubic cell at the isotropic Ti site is 0.02%, so any dichroism elsewhere is physical rather than numerical.

For the tetragonal cell with its polar axis along the beam, the apical-oxygen O K edge is strongly dichroic, with $\max|\Delta|/\max S = 78\%$ and $\int|\Delta|/\int S = 47\%$ (Fig. 5). The origin is a σ/π anisotropy: with $\mathbf{q} \perp c$ the edge shows a sharp π^* peak ≈ 8.5 eV above onset from O $2p_{x,y}$, whereas with $\mathbf{q} \parallel c$ it shows σ^* features at ≈ 11 and 18.5 eV from the O $2p_z$ states directed along the Ti–O–Ti chain. The excited site here is the apical oxygen, which lies on that chain parallel to the polar axis. The edge of the cell as a whole is the multiplicity-weighted sum of one apical and two equatorial oxygen, and the equatorial site behaves very differently: its own chain lies in-plane, and it is only 22% dichroic. Carrying twice the multiplicity, it brings the weighted full O K edge to 37% rather than 78%. The two equatorial atoms are equivalent for $\mathbf{q} \parallel c$ but not for $\mathbf{q} \perp c$, so the second orientation was computed explicitly as a third momentum-transfer direction rather than estimated. One systematic remains, the chemical shift between the two oxygen sites, which these calculations do not fix; sweeping it over ± 1 eV moves the weighted edge over 22–37%, while the integral metric is more stable at 16–23%. The rotational-invariance test passes essentially exactly: mapped into the crystal frame the $c \parallel z$ and $c \parallel x$ cells agree to 0.02–0.03%, the floor set by the cubic null test, so the anisotropy is physical rather than an artefact of the momentum-transfer machinery.

A large orientation dichroism does not by itself constitute a polarisation measurement, since the tetragonal backbone is anisotropic even without off-centring. Holding the lattice parameters fixed and varying only the internal displacement s separates the two contributions (Fig. 6). At $s = 0$, corresponding to zero polarisation with full tetragonal strain, the dichroism is already 70.1%, rising to 78.2% at $s = 1$. Approximately 70 of the 78 points are therefore the strain-induced backbone anisotropy, and the signature of the off-centring itself is the remaining monotonic contribution: an 18.7% change in the along-beam-probed O K spectrum between $s = 0$ and $s = 1$, 4.3% at $s = 0.5$, together with 8 points of additional dichroism. This is the first-principles counterpart of the empirical observation that the O K edge responds to titanium off-centring [44], and it renders the measurement a polarisation probe rather than merely a strain probe.

The intrinsic dichroism must then survive the acquisition geometry. For the O K edge at 300 keV, $\theta_E = 1.09$ mrad and the magic angle is 4.32 mrad $= 3.98\,\theta_E$, reproducing the textbook factor of ≈ 4 and validating the aperture model. Folding the anisotropy through the collection geometry gives a surviving fraction of 0.62 at $\beta = 1$ mrad, 0.28 at 2 mrad, -0.04 at 5 mrad and -0.33 at 100 mrad. A combined ptychography and EELS experiment using one pixelated detector with a central hole would collect at $\beta \approx 100$ mrad and would therefore suppress the signal it was built to measure.

That calculation is nevertheless the textbook one, which assumes parallel illumination, and it is the only geometry for which a magic angle is defined. A focused probe is not that geometry. The momentum transfer is the vector difference of the incident and scattered transverse angles, so a convergence semi-angle α spreads $\mathbf{q}$ on its own, irrespective of how small the spectrometer aper-

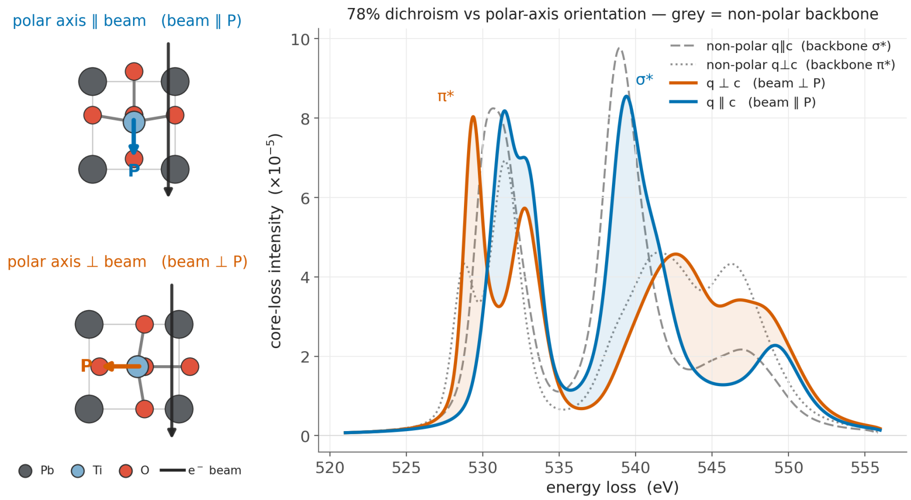

FIG. 5. Orientation dichroism of the $PbTiO_3$ apical-oxygen O K edge from core-hole CASTEP and OptaDOS. With the momentum transfer across the polar axis ($\mathbf{q} \perp c$, beam $\perp$ $\mathbf{P}$) the edge shows a sharp π^* peak; along it ($\mathbf{q} \parallel c$, beam $\parallel$ $\mathbf{P}$) it shows the σ^* features of the O $2p_z$ states directed along the Ti–O–Ti chain. Grey curves are the same two orientations for the zero-polarisation, non-polar and equally strained reference, isolating the backbone contribution.

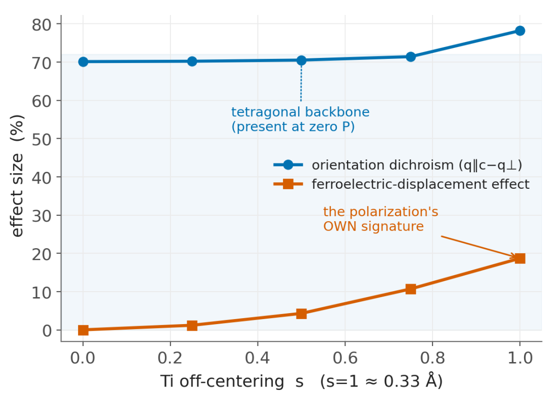

FIG. 6. Separating strain from polarisation. Holding the lattice parameters fixed and varying only the internal Ti off-centring s, the total orientation dichroism (blue) is already 70% at zero polarisation, corresponding to the tetragonal backbone, whereas the signature of the ferroelectric displacement itself (orange) grows monotonically to 19% at the experimental off-centring $|\mathbf{u}| \approx 0.33\,\text{Å}$.

ture is. Averaging the dipole weights over both cones changes the picture. At $\beta = 1\,\text{mrad}$ the surviving fraction falls from 0.62 under parallel illumination to 0.29 at $\alpha = 2\,\text{mrad}$ and to -0.04 at $\alpha = 5\,\text{mrad}$. Closing the collection aperture is therefore not by itself a remedy, and recovering the *full* anisotropy would require illumination as parallel as $\theta_E = 1.09\,\text{mrad}$, which is incompatible with a focused probe.

The consequence for a combined instrument is more favourable than that reads, because the anisotropy does not vanish beyond the magic angle. It inverts and then saturates, approaching about a third of its intrinsic value. At the $\alpha = 100\,\text{mrad}$ convergence that atomic-resolution ptychography requires, the surviving fraction is -0.32 to -0.34 for every collection angle between 1 and 100 mrad. It is, in other words, flat: the spectrometer aperture is no longer a design variable, because the convergence already sets the averaging. Folding the measured weighted edge through this geometry, a domain with its polar axis along the beam differs from one with the axis in-plane by 11.9% in the collected O K edge, which needs $\approx 1.3 \times 10^3$ counts per channel for SNR 3 rather than the 4.5×10^4 implied by a 2% anisotropy.

This is the quantitative case for a hollow detector. Placing the spectrometer entrance in the centre of the pixelated detector and reserving the outer annulus for imaging costs the spectroscopy nothing that a smaller aperture would recover, while collecting the entire bright-field disc gives it the maximum available signal. The hole radius may be chosen to suit the imaging.

Against this stand limits that no choice of aperture removes. The dipole ELNES is even in $\mathbf{q}$, so it measures the magnitude of the along-beam polarisation but not its sign. A dynamical multislice core-loss simulation of up- and down-polarised domains confirms as much, the two giving O K spectra identical to within 1.0%. In the labyrinth itself, viewed along this zone axis, the polarisation is in any case mostly in-plane: over 2560 polar titanium the median angle from the beam is 82° (interquartile range 73–87°), an along-beam fraction of only ≈ 0.14.

That last point does not, however, make the computed case an upper bound that a real measurement would fall far below. Repeating the calculation on a cell built at the labyrinth's median 82° tilt gives a dichroism of 76.3%, against 78.2% for the fully aligned case. What changes is not the magnitude but the character, and exactly so: with the polar axis along the beam, $\mathbf{q} \parallel$ beam gives the

σ^* feature at 18.7 eV and $\mathbf{q} \perp$ beam the π^* at 8.6 eV, whereas with the axis in-plane the same two features appear at the same two energies with their assignments exchanged. The edge reports the *orientation* of the polar axis relative to the beam, the sense of the asymmetry identifying which way it lies, rather than the magnitude of the along-beam component. A measurement on the labyrinth would therefore return a large dichroism reporting an in-plane axis, not an absence of signal. It remains true that only a small part of the effect belongs to the off-centring rather than to the tetragonal backbone, and that the dipole limit forbids the sign.

IV. CONCLUSIONS AND FURTHER WORK

Fitting a reconstructed multislice-ptychographic volume as a superposition of a measured single-atom response recovers the oxygen sublattice of a $PbTiO_3/SrTiO_3$ superlattice where the established deconvolve-then-detect workflow does not, at 95.5% bulk recall against at most 68% over the same band. The whole of that margin lies in the axially overlapped oxygen (82% against 9–26%). It is not a tuning artefact, since the limitation of deconvolution is the axial spatial frequencies the aperture never records. The method also returns species labels (1.1% confusion) and calibrated per-atom uncertainty, neither of which a peak detector provides.

The uncertainty machinery is arguably the more transferable contribution. A joint Cramér–Rao bound recovers the twofold variance inflation that a per-peak bound misses wherever responses overlap, and Mondrian conformal calibration converts it into intervals holding their nominal coverage per stratum. These contain no hand-tuned constant and transfer unchanged to an independent reconstruction, on which hand-tuned error floors give only 38–56% coverage against a 68% target. Reporting coverage rather than a fitted σ is therefore to be encouraged: the same pipeline that reads σ-calibrated at one dose reads 38% species confusion at another, while its precision still reads 0.95.

Propagated into the physics, the result cuts both ways. The in-plane polarisation map is quantitative, with a median error of 0.007 Å in magnitude and 0.9° in direction against a 0.27 Å signal. The along-beam component, by contrast, is not measured at all, because differencing two 0.3 Å depth coordinates cannot resolve a 0.07 Å displacement. The calibrated uncertainty establishes this without ground truth, which is the purpose of such machinery. First-principles core-hole DFT indicates that the missing component is spectroscopically accessible in principle, the apical-oxygen O K edge being 78% dichroic to the polar-axis orientation, of which a monotonic 19% is the signature of the displacement rather than of the tetragonal backbone. Weighted over both oxygen sites the full edge retains 37% of that dichroism. The dipole limit nevertheless forbids recovery of the sign, and the magic angle forbids use of the 100 mrad ptychography aperture for the purpose.

This leaves clear openings for further work. The assumption of one kernel for all species is weakest where it matters most, since the larger Debye–Waller factor of oxygen broadens its response under thermal scattering; in-situ vacancy-difference kernels would supply an in-crystal oxygen matched filter. The 5% cage-composition tail of Sec. III E is a detection rather than a localisation problem and requires a calibrated cage-completeness criterion, while learned denoising before fitting [47] is a natural route below $10^6\,\mathrm{e\,Å^{-2}}$. The weighted O K edge carries a 22–37% systematic from the inter-site chemical shift, which a separate no-hole calculation would fix. Beyond that, a faithful vortex simulation would inject a per-atom $\sigma(E)$ keyed to each oxygen's local polar-axis orientation. The clearest instrumental conclusion is that a hollow detector is the right geometry, for a reason that is not the obvious one. A convergent probe spreads the momentum transfer by itself, so beyond the magic angle the anisotropy inverts and saturates near a third of its intrinsic value rather than vanishing, and at the convergence imaging requires it is independent of the collection angle to within a few percent. Nothing is therefore lost by giving the spectrometer the central disc and the imaging the outer annulus. The limit on such an experiment in this material is not the detector but the specimen: the polar axes of the labyrinth are clustered about the in-plane direction, so the orientation contrast between neighbouring domains is of order 1% rather than the 12% available between the limiting orientations, and a zone axis presenting a wider spread of axes would serve far better. The open question is whether the two can be fused into a single three-dimensional vector polarisation measurement. Energy-filtered ptychography of the oxygen sublattice [11] and momentum-selective EELS of these flux-closure structures [48] both suggest routes.

Stage	Parameter	Value
Sample	composition	$PbTiO_3/SrTiO_3$, 19 440 atoms
	supercell	$70.0 \times 70.0 \times 47.9$ Å
Probe	energy / α	300 keV / 100 mrad
	defocus	-20 Å (20 Å overfocus)
Scan	window / step	20 Å / 0.10–0.15 Å
Sim.	slice thickness	2.0 Å (0.5 Å fine)
	frozen phonons	16 configs; per-species B
	detector	200 mrad, binned to 356^2 px
	dose	10^4–10^{10} e Å^{-2} (Poisson)
Recon	engine	PtychoShelves LSQ-ML
	layers N_L	70 / 105 ($\Delta z_{\rm rec} = 1.0$ / 0.67 Å)
	regularisation	inter-layer off; $\beta_{\rm LSQ} = 0.1$
DFT	code / functional	CASTEP [39] / PBE
	core hole	full ($1s^1/2p^5$), charge $+1$
	supercell	$2 \times 2 \times 2$ (40 atoms)
	spectra	OptaDOS [41], $\hat{\mathbf{q}}$-resolved

TABLE A1. Simulation, reconstruction and first-principles parameters.

Appendix A: Simulation and reconstruction parameters

Table A1 collects the parameters of the forward simulation, the reconstruction and the first-principles calculations. Reproducing the reconstruction from the simulated data also depends on the geometry conventions recorded here. The object pixel is set by the diffraction angle rather than by the scan step, giving $\mathrm{d}x = 0.0492$ Å for the binned 356^2 detector, and the detector requires a single transpose relative to the PtychoShelves convention. A two-stage schedule, a down-sampled 178-pixel pre-solve followed by a full-resolution 356-pixel refinement, accelerates convergence of the non-convex objective.

Appendix B: Localisation algorithm

The five stages below operate on the reconstructed phase volume and the measured kernel alone. No ground-truth quantity enters at any point.

Preprocessing. Each layer has its median subtracted, so that vacuum maps to zero, and a heavily Gaussian-blurred copy of the layer is removed to suppress the slowly varying channelling haze that otherwise biases weak oxygen amplitudes. The result is clipped at zero and the entrance and exit artefact layers ($z < 2$ Å, $z > 66$ Å) are trimmed.

Column seeding. In-plane the problem is well conditioned, the kernel being two pixels wide, so local maxima of the depth-averaged image over a 1.4 Å non-maximum window give ≈ 98 column seeds across the field.

Support selection. Around each seed a fixed tube is taken, ± 0.5 Å in-plane and spanning the full interior depth, and matching pursuit is run within it with the unit-norm kernel: the residual is correlated with K, the global maximum is accepted as an atom, βK is subtracted, a neighbourhood of ± 1.5 layers by ± 0.2 Å is excluded, and the process repeats. Iteration stops at $\max(K \star R) < k\,\sigma_{\rm MF}$, where $\sigma_{\rm MF}$ is a median-absolute-deviation estimate of the matched-filter noise taken over the sub-median voxels of the same tube and $k = 2$. Because each atom carries its own (x, y, z), the ferroelectric lean of a column costs nothing.

Joint refinement. Two sweeps of Gauss–Newton refinement follow. For atom i the models of all other atoms are subtracted and (β, x, y, z) is fitted on the local patch by linearising K, with $\partial K/\partial x$ and its counterparts obtained by central differences on the measured kernel, giving sub-voxel positions. A candidate is rejected if the normalised correlation between its patch and βK falls below 0.5. This is a shape test rather than an amplitude cut, which matters because an amplitude cut cannot separate faint oxygen from spurious detections whereas a shape test can.

Species assignment and lattice-constrained detection. Columns self-classify by k-means on their own per-column amplitude statistics, which are well separated in this material: the 75th percentile is ≈ 4.1 for lead columns, ≈ 1.5 for mixed B–O columns and ≈ 0.66 for pure-oxygen columns, with no band overlap. On B–O columns titanium and oxygen alternate every ≈ 1.95 Å, and species is assigned by per-column amplitude with a dead zone, ambiguous atoms being resolved by local parity, that is by the distance of $z_i - z_j$ modulo the fitted period to confident neighbours. Parity is deliberately translation-invariant, because a global comb phase accumulates period error towards the column ends and inverts labels there. Expected sites are then enumerated by extending $\pm kp$ from each found atom of the same species, and $\pm p/2$ from each titanium for oxygen; where a site is empty, meaning further than 0.8 Å from any same-species atom, a matched-filter search is run on the residual within gates of ± 0.7 Å in depth and ± 0.35 Å in-plane, followed by refinement. Such a detection is accepted only if its fit quality clears a reduced bar of 0.35 and its amplitude is lattice-consistent, lying between 0.45 and 1.6 times the column's same-species median. The reduced bar is permitted for oxygen alone, the position prior substituting for the missing contrast; the same treatment applied to lead and titanium produced mislocated atoms with overconfident errors and was abandoned. A post-fit occupancy guard rejects any constrained detection landing within 1.2 Å of an atom already accepted in the same tube, no two atoms in the reference structure being closer than 1.70 Å. All such atoms carry a flag through to the exported coordinates, so that the lattice prior is never silently absorbed.

sep.	pair	VIF(β)	VIF(z)	σ_z low by
3.90 Å	Pb–Pb, Ti–Ti	1.04	1.04	×1.02
1.95 Å	Ti–O (apical)	2.28	2.02	×1.42
0.99 Å	near-degenerate	13.3	3.33	×1.83

TABLE C1. Variance inflation of the joint fit relative to the per-peak bound, measured on the kernel used here. Every apical oxygen sits at the 1.95 Å separation.

Appendix C: Uncertainty quantification

1. Joint versus per-peak Cramér–Rao bound

For the stacked parameter vector $\theta = (\theta_1, \ldots, \theta_N)$, with $\theta_i = (\beta_i, \mathbf{r}_i)$, the Fisher information of Eq. (3) under homoscedastic residuals is $\mathbf{F} = \mathbf{J}^\top \mathbf{J}/s^2$, and the attainable variance of θ_i is the corresponding diagonal block of $\mathbf{F}^{-1}$ rather than the inverse of the diagonal block $\mathbf{F}_{ii}$. The two coincide only for an orthogonal design; in general $[\mathbf{F}^{-1}]_{ii} \geq [\mathbf{F}_{ii}]^{-1}$, and the ratio is the variance inflation factor (VIF) of the overlapping templates. Evaluating $[\mathbf{F}_{ii}]^{-1}$, in which each atom is fitted with its neighbours held fixed, is the per-peak bound used to predict single-column and single-dopant precision [6, 17]; on a dense sublattice it is the bound obtained with all neighbours known exactly. Table C1 gives the inflation measured for our kernel. Since 85% of atoms in the field have a neighbour within 2.5 Å, the per-peak form is optimistic almost everywhere. We therefore assemble $\mathbf{J}$ over all atoms in a subvolume and invert $\mathbf{F}$ once, a $4N \times 4N$ system with $N \leq 45$ and of negligible cost, taking each atom's marginal variance. This restores the inflation per atom automatically and without a tuned constant.

The systematic term σ_{sys} is obtained by generating a synthetic response with one measured kernel, refitting it with another admissible kernel, and taking the resulting position spread. Being computable without ground truth, it transfers to experimental data. On the coherent reconstruction, where the lead and titanium kernels are nearly identical, it is small ($\sigma_z = 0.02$ Å); it grows on datasets whose kernels genuinely differ.

2. Conformal calibration and measured coverage

Nonconformity scores $s = |\Delta|/\sigma$ are formed per axis and the empirical $(1-\alpha)$ quantile taken with the finite-sample correction, the reported half-width being $q_{1-\alpha}\sigma$. Coverage is guaranteed only marginally, and the error is strongly heteroscedastic across the lattice, so calibration is performed per stratum, the strata being species × detection mode × depth band. Table C2 gives the measured held-out coverage on the coherent reconstruction and on the independent dosed volume. On the latter the conformal step is re-run on that volume's own matched atoms, which is the intended workflow, since applying one volume's quantile table to another assumes an exchangeability that fails when registration quality differs. The per-stratum quantiles $q_{68}(z)$ span 6.7–14.4 on the coherent volume and 10.9–23.3 on the dosed one, the wider spread on the harder volume being precisely the regime a single floor cannot serve.

volume	target	coverage ($x/y/z$)	tuned floors
coherent	68%	69/69/69%	—
coherent	95%	96/96/96%	—
dosed 10^{10}	68%	69/69/69%	38/47/56%
dosed 10^{10}	95%	96/96/96%	—

TABLE C2. Held-out coverage of the calibrated intervals, split-conformal on a 50/50 split. The final column gives the hand-tuned per-species error floors used in an earlier version of this work, calibrated to 68% on the coherent volume and applied unchanged to the dosed one.

Exported per atom are the model σ, the 95% conformal half-width taken as the default, and a within-column amplitude posterior p_{species} for the assigned label. The last is a relative confidence rather than a calibrated probability, and it is undefined in the useful sense on single-species columns, but atoms with $p_{\text{species}} < 0.9$ are enriched for mislabelling by roughly 25× over the 1.1% base rate.

Appendix D: Kernel provenance and noise robustness

The kernel is measured by propagating a single isolated atom through the identical simulation and reconstruction pipeline. Isolated light atoms do not survive this procedure: the titanium and oxygen kernels extracted from the thermally averaged, dosed geometry have signal-to-noise ratios of 2.4 and 2.7 respectively, with their interior maxima displaced to the volume corner, and are unusable. The lead kernel of the coherent reconstruction is by contrast clean, with a signal-to-noise ratio of 81.5 and an axial full width at half maximum of 1.0 Å, and the corresponding thermally averaged lead kernel reaches 23.6 at 2.0 Å. A kernel derived instead from lead blobs within the crystal is measurably worse, its axial width inflated to 4.0 Å by unresolved neighbours, and it reduces lead recall from 85% to 65%. The lead shape is therefore used for all species. The principal known limitation of this choice is that the Debye–Waller factor of oxygen exceeds that of titanium ($B = 0.80$ against 0.45 Å^2), so that under thermal scattering the true oxygen response is genuinely broader than the lead response, and the assumption weakens exactly where it matters most.

Table D1 reports a controlled noise sweep in which Gaussian noise is injected into the reconstructed volume with geometry, kernel and calibration held fixed. The intrinsic vacuum noise floor of the volume is $\sigma \approx 0.022$ and the oxygen peak amplitude ≈ 0.06, so the collapse of oxygen recall between $\sigma = 0.01$ and $\sigma = 0.02$ corre-

injected σ	found	precision	Pb	Ti	O	O overlapped
0	1834	0.97	88 %	89 %	89 %	82 %
0.01	1793	0.97	88 %	91 %	86 %	74 %
0.02	1421	0.98	87 %	90 %	57 %	37 %
0.04	795	0.96	87 %	86 %	5 %	18 %
0.08	408	0.95	87 %	0 %	0 %	0 %

TABLE D1. Injected-noise sweep on the coherent volume, all other settings fixed. Recalls are over the full depth rather than the bulk band. Species confusion rises from 1.1% at $\sigma = 0$ to 3.7% at 0.02, 58.7% at 0.04 and 74.6% at 0.08. The final column is the axially overlapped oxygen of Sec. III B, the population on which the method's advantage rests, and it is also the population the noise removes first.

sponds to the oxygen signal falling below the noise. It is not a thresholding artefact and no choice of detection floor removes it. The sweep also demonstrates the insensitivity of precision as a diagnostic: at $\sigma = 0.08$ the recall of both titanium and oxygen is zero and three in four species labels are wrong, yet precision still reads 0.95, because the surviving bright lead atoms remain correctly placed. Species confusion and σ-coverage are the quantities that respond, and the failure is at least conservative, the lattice-consistency amplitude gate causing constrained oxygen detections to abstain rather than fabricate.

Appendix E: Reproducibility protocol

All code, the input structure and the per-run parameters are held in the project repository, the pipeline being scripted end to end from simulation (`sim/simulate_4dstem.py`) and reconstruction (PtychoShelves `GPU_MS`) through localisation (`atomfind/run_atomfind.py`) and polarisation (`atomfind/polarisation.py`) to the DFT staircase (`eels/build_cells.py`, `run_coreloss.slurm`, `analyze_elnes.py`), with a unit-test suite guarding structure generation and parsing.

1. The nominated result

The result offered for independent reproduction is the localisation-with-uncertainty output of Sec. III E. It is packaged as a standalone repository containing the localisation code alone, together with a checksummed archive of four inputs: the reconstructed phase volume as a $70 \times 404 \times 404$ complex array, the measured lead and titanium kernels, and the reference structure used for scoring, supplied both as deposited and pre-transformed into the beam frame. The simulation and reconstruction that generate the volume require GPU resources well beyond a few hours and are therefore supplied pre-computed; everything downstream of them is reproduced from scratch. The whole procedure is

```
python3 -m venv venv && . venv/bin/activate
pip install -r atomfind/requirements.txt
tar xzf atomfind_data_v1.tar.gz -C atomfind/data/
python atomfind/run_atomfind.py \
       --preset NL70_coherent --out ./out
python atomfind/polarisation.py --out ./out
```

executed from the directory containing `atomfind/`. The run requires a few minutes on a single core and no GPU, and depends only on NumPy, SciPy, h5py, Matplotlib and ASE. No path is hard-coded: the data files are located by name, through an environment variable, a directory adjacent to the package, or the `--data-dir` option.

These instructions were executed rather than merely written. The package was rebuilt from the repository, unpacked into a bare virtual environment holding no scientific software beyond those five libraries, and run end to end. It reproduced every quantity below, the largest relative discrepancy anywhere in the emitted record being 1.3×10^{-11}, on a mean null-site amplitude that accumulates many small terms. That environment ran NumPy 2.0 against the development environment's NumPy 1.26, which is the more useful statement: the agreement is across library generations and not merely across two copies of one installation.

2. Expected output and its uncertainty

The first command writes `found_atoms.csv`, containing for each atom its element, its (x, y, z) in ångström in the reference frame, the per-axis 95% conformal half-widths, the model σ, the species confidence and a flag marking lattice-constrained detections, together with the conformal quantile table. The second writes the Ti–O_6 off-centring with its propagated uncertainty. On the same input the run is deterministic and should reproduce 1834 atoms with precision 0.97, bulk recall 95.6/96.4/95.5% for Pb/Ti/O, species confusion 1.1%, in-plane RMS accuracy 0.032 Å and depth RMS 0.37 Å.

The uncertainty attached to each reported coordinate is the exported conformal half-width, and the statement that validates it is the coverage figure itself: 96% of true errors fall within the nominal 95% interval, per stratum, verified held out on a 50/50 split (Table C2). Typical half-widths are 0.02 Å in-plane and 0.5 Å in depth, rising to 1.5 Å in depth for weak oxygen near the exit surface. Because they are strongly heteroscedastic, the per-atom interval and not a global figure is the quantity to propagate. For the derived polarisation the in-plane off-centring carries a propagated σ of 0.011 Å and the along-beam component 0.237 Å.

3. Tolerances and known failure modes

The single stochastic element is the conformal calibration split, which is seeded; re-seeding changes the per-stratum quantiles by less than 5% and leaves the coverage figures within one percentage point. Coverage is a finite-sample quantity, so strata containing fewer than about twenty atoms, of which the constrained-oxygen entrance band is the example here, show conformal noise of several percentage points and should not be read as failures.

The scripts report their own failure modes rather than absorbing them, and a peer should expect to see both. Approximately 28% of bulk titanium do not acquire a complete oxygen cage, and these are excluded from the polarisation map rather than completed from the lattice. A further 5% have a cage of the wrong composition, for which the in-plane error rises to a median of 0.45 Å while the positional interval covers only 12% of them. This second mode is a detection failure rather than a localisation failure, and is visible in the exported cage completeness and species confidence rather than in the coordinate error bars.

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